3 EXAM PRACTICE

1 Which of the following statements about the properties associated with ionic and covalent compounds is correct?

A An ionic compound cannot undergo electrolysis.

B The only covalently bonded substances with high melting temperatures are those in which hydrogen bonds are present.

C A compound cannot contain both ionic bonding and covalent bonding.

D Ionic compounds differ from metals because they do not conduct electricity in the solid state. [1]

(Total for Question 1 = 1 mark)

2 Which is the best explanation for the ability of graphite to act as a solid lubricant?

A It has delocalised electrons that are able to flow parallel to the layers.

B The covalent bonds within the layers are strong.

C Gas molecules are trapped between the layers.

D The forces of attraction between the layers are very weak. (1)

(Total for Question 2 = 1 mark)

3 Which of the following molecules contains six bonding electrons?

A C_2H_4 **B** CO_2 **C** H_2S **D** NCl_3 [1]

(Total for Question 3 = 1 mark)

4 A solid melts just above 100 °C. It does not conduct electricity even when molten.

Which of the following structures is the compound most likely to have?

A giant ionic

B giant covalent

C simple molecular

D giant metallic [1]

(Total for Question 4 = 1 mark)

5 The species Ar, K^+ and Ca^{2+} are isoelectronic.

In what order do their radii increase?

	smallest ⟶		largest
A	Ar	Ca^{2+}	K^+
B	Ar	K^+	Ca^{2+}
C	Ca^{2+}	Ar	K^+
D	Ca^{2+}	K^+	Ar

[1]

(Total for Question 5 = 1 mark)

6 Which of the following molecules does not have a permanent dipole?

A CF_4 **B** CHF_3 **C** CH_2F_2 **D** CH_3F [1]

(Total for Question 6 = 1 mark)

7 Lithium, magnesium and sodium all exhibit metallic bonding.

(a) State what is meant by metallic bonding. [2]

(b) State how a metal can conduct electricity. [1]

(c) Explain why magnesium is a better conductor of electricity than sodium. [2]

(d) Predict whether or not sodium is a better electrical conductor than lithium. Justify your answer. [2]

(Total for Question 7 = 7 marks)

8 The table shows some of the properties of four substances, **A**, **B**, **C** and **D**.

State the type of bonding and structure that is likely to be present in each of the substances.

In each case justify your answer. [8]

Substance	Solubility in water	Electrical conductivity		Melting temperature / °C
		of solid	in aqueous solution	
A	insoluble	poor	—	1610
B	soluble	poor	good	801
C	insoluble	good	—	1083
D	soluble	poor	fair	–78

(Total for Question 8 = 8 marks)

9 The electron pair repulsion (EPR) theory can be used to predict the shape of simple molecules and ions.

(a) State the main assumptions of the EPR theory. [3]

(b) Draw a dot-and-cross diagram for each of the following molecules/ions.

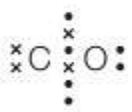
(i) BF_3 (ii) NH_3 (iii) NH_4^+ (iv) SF_6 [4]

(c) Predict the shape and bond angles in each of the molecule/ions in part (b). [5]

(Total for Question 9 = 12 marks)

10 Nitrogen and carbon monoxide are both gases consisting of diatomic molecules. Both gases are colourless, odourless and tasteless, but unlike nitrogen, carbon monoxide is extremely toxic.

(a) The dot-and-cross diagram for carbon monoxide,, showing only the outer electrons, is:



Copy the diagram and add labels to identify:

(i) a lone pair of electrons

(ii) a covalent bonding pair of electrons and

(iii) a dative covalent bonding pair of electrons. [3]

(b) The nitrogen and carbon monoxide molecules are isoelectronic.

(i) State what is meant by isoelectronic. [1]

(ii) The HCN molecule is also isoelectronic with N_2 and CO.

Draw a dot-and-cross diagram for HCN, showing only the outer shell electrons. [2]

(iii) Suggest why nitrogen is much less reactive than either carbon monoxide or hydrogen cyanide. [1]

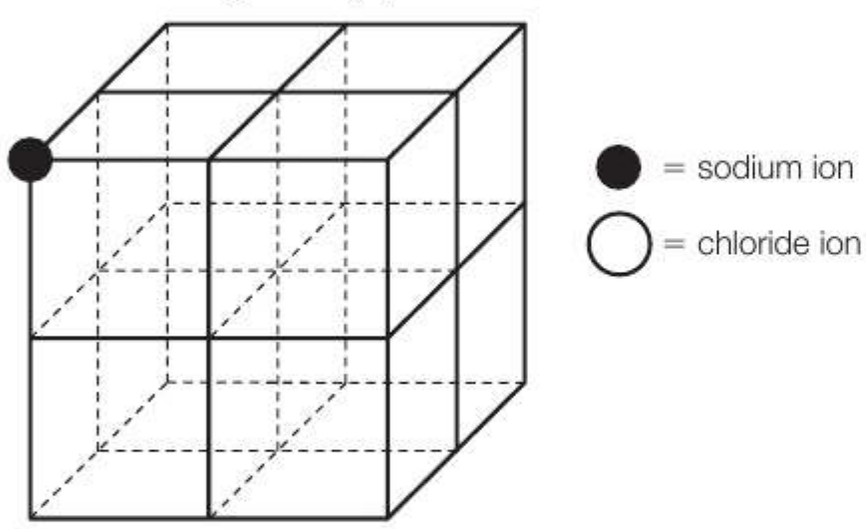
(Total for Question 10 = 7 marks)

11 Sodium chloride is an ionic compound containing sodium ions (Na^+) and chloride ions (Cl^-).

It has a fairly high melting temperature, is soluble in water and is a poor conductor of electricity when solid, but good when molten.

(a) Draw dot-and-cross diagrams to show the changes in arrangement of electrons that take place when sodium (Na) reacts with chlorine (Cl_2) to form sodium chloride. Show only the outer electrons in each case. [4]

(b) Complete the diagram to show the structure of sodium chloride using the key provided. [2]



(c) Explain why sodium chloride

(i) has a fairly high melting temperature [3]

(ii) is soluble in water [3]

(iii) is a poor conductor of electricity when solid, but a good conductor when molten. [2]

(Total for Question 11 = 14 marks)

12 This question is about four simple molecular compounds: water, ammonia, methane and boron trichloride.

(a) Complete the table to show the number of bonding pairs and non-bonding (lone) pairs of electrons in one molecule of each compound. [4]

Formula of molecule	H_2O	NH_3	CH_4	BCl_3
Number of bonding pairs				
Number of non-bonding pairs				

(b) The ammonia molecule and the boron trichloride molecule both contain polar bonds.

(i) State what is meant by a polar bond, and explain how the polarity arises. [3]

(ii) Explain why the ammonia molecule is polar, but the boron trichloride molecule is not. [3]

(Total for Question 12 = 10 marks)